25900083 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCFTQ100 - BASIC BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. With the help of structure, list the properties of insoluble component of starch [A] / [CO1]
2. What does the secondary structure of proteins refer to? [U] / [CO2]
3. What is halogenation of fats? Discuss its importance. [A] / [CO4]
4. Define and give the importance of Kiliani-Fischer synthesis [U] / [CO1]
5. How does enzyme perform its catalytic role? [U] / [CO3]
6. Classify fatty acids based on their importance in diet. [An] / [CO4]
7. What are Ligases? Give examples. [K] / [CO3]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. Make a detailed note on the following [U] / [CO1]

   a. Mutarotation
   b. OpticalIsomerism
   c. D and L Isomers
   d. Chirality and 2^n rule

9. With the help of a graph, explain how temperature affects enzyme activity. [U] / [CO3]

10. What are Lipids? Narrate the functions of lipids. [U] / [CO4]

11. Elucidate the structure and properties of D-Galactose [U] / [CO1]

12. Elaborate in detail on the physical properties of amino acids. [U] / [CO2]

13. With the help of a structure, explain the properties and functions of cholesterol. [U] / [CO4]

14. Enumerate the chemical reactions of amino acids due to NH_3 group. [K] / [CO2]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

15. Explain different types of reversible inhibition occuring in an enzyme catalysed reaction. [A] / [CO3]

16. Describe in detail on classification of carbohydrates. [U] / [CO1]

17. Write an essay on the structural organisation of proteins. Add a note on the chemical bonds responsible for its formation. [U] / [CO2]

18. Elaborate in detail on the classification of fatty acids with examples. Make a note on the structure and functions of essential fatty acids. [U] / [CO4]

[10x2 = 20]